

Bulan 2: Django Fundamentals

Minggu 1: Django Setup & Basic Concepts

Hari 1: Django Installation & Project Structure

Project: StartupKit

- Deskripsi: Django boilerplate generator dengan best practices structure
- Fitur: Custom user model, settings split (dev/prod), gitignore template
- Keunikan: Interactive setup wizard untuk pilih tech stack

Hari 2: Django Apps & MVT Pattern

Project: QuickNotes

- Deskripsi: Note-taking app sederhana untuk memahami MVT
- Fitur: Create, read, update, delete notes dengan rich text
- Keunikan: Auto-save draft setiap 30 detik

Hari 3: URL Routing & Views

Project: LinkShortly

- Deskripsi: URL shortener dengan custom aliases dan analytics
- Fitur: Generate short links, click tracking, QR code
- Keunikan: Vanity URLs dan expiration dates

Hari 4: Templates & Template Tags

Project: BlogCanvas

- Deskripsi: Blog engine dengan template inheritance dan custom tags
- Fitur: Post categories, reading time estimator, related posts
- Keunikan: Dynamic template switcher (themes)

Hari 5: Static Files & Media

Project: PhotoDrop

- Deskripsi: Image upload dan gallery dengan thumbnail generation

- Fitur: Drag-drop upload, auto-resize, EXIF data display
- Keunikan: Lazy loading dan progressive image loading

Hari 6: Django Admin Basics

Project: AdminPro

- Deskripsi: Enhanced admin interface dengan custom actions
- Fitur: Bulk operations, export CSV/Excel, advanced filters
- Keunikan: Admin dashboard dengan stats widgets

Hari 7: Models & Database Fields

Project: InventoryPro

- Deskripsi: Inventory management dengan berbagai field types
- Fitur: Product catalog, stock tracking, low stock alerts
- Keunikan: Barcode generator dan scanner integration

Minggu 2: Advanced Models & ORM

Hari 8: Model Relationships (ForeignKey)

Project: TeamSync

- Deskripsi: Team management app dengan hierarchical structure
- Fitur: Teams, members, roles, permissions
- Keunikan: Organizational chart visualization

Hari 9: Many-to-Many Relationships

Project: CourseHub

- Deskripsi: Online course platform dengan student enrollment
- Fitur: Courses, students, instructors, prerequisites
- Keunikan: Learning path suggerer

Hari 10: Model Methods & Properties

Project: FitnessLog

- Deskripsi: Workout tracker dengan calculated metrics

- Fitur: Exercise log, BMI calculator, calorie burn estimator
- Keunikan: Auto-generate workout plans based on goals

Hari 11: QuerySets & ORM

Project: DataDash

- Deskripsi: Analytics dashboard dengan complex queries
- Fitur: Filtering, aggregation, annotations, subqueries
- Keunikan: Query builder UI untuk non-developers

Hari 12: Model Managers

Project: ContentStream

- Deskripsi: Content management dengan custom querysets
- Fitur: Published/draft states, scheduled publishing, archiving
- Keunikan: Time-travel feature (view site at past date)

Hari 13: Database Migrations

Project: MigrateHelper

- Deskripsi: Migration management tool dengan visualization
- Fitur: Migration graph, squashing, conflict detection
- Keunikan: Safe rollback dengan data preservation

Hari 14: Model Validation

Project: FormGuard

- Deskripsi: Advanced form dengan custom validators
- Fitur: Cross-field validation, async validation, error handling
- Keunikan: Real-time validation dengan AJAX

Minggu 3: Forms & User Input

Hari 15: Django Forms

Project: SurveyMaker

- Deskripsi: Survey builder dengan dynamic form generation
- Fitur: Multiple question types, conditional logic, analytics
- Keunikan: A/B testing untuk survey questions

Hari 16: ModelForms

Project: ProfileCraft

- Deskripsi: User profile editor dengan image cropping
- Fitur: Bio, skills, social links, avatar upload
- Keunikan: LinkedIn-style completeness indicator

Hari 17: Form Validation & Cleaning

Project: SecureRegister

- Deskripsi: Registration system dengan advanced validation
- Fitur: Password strength, email verification, spam detection
- Keunikan: Passwordless magic link option

Hari 18: Formsets

Project: InvoiceBuilder

- Deskripsi: Invoice generator dengan line items
- Fitur: Multiple items, tax calculation, PDF generation
- Keunikan: Recurring invoice automation

Hari 19: File Uploads

Project: DocVault

- Deskripsi: Document management dengan versioning
- Fitur: File upload, preview, sharing, access control
- Keunikan: Virus scanning dan file type validation

Hari 20: AJAX Forms

Project: LivePoll

- Deskripsi: Real-time polling app dengan instant results

- Fitur: Create polls, vote, live results chart
- Keunikan: No page reload, WebSocket updates

Hari 21: Form Widgets Customization

Project: DateTimePro

- Deskripsi: Event scheduler dengan custom datetime picker
- Fitur: Recurring events, timezone support, reminders
- Keunikan: Natural language date input ("next friday 3pm")

Minggu 4: Authentication & Authorization

Hari 22: User Authentication

Project: AuthFlow

- Deskripsi: Complete auth system dengan login/logout/register
- Fitur: Remember me, password reset, account activation
- Keunikan: Social login preview (Google, GitHub)

Hari 23: User Registration & Profiles

Project: MemberZone

- Deskripsi: Membership site dengan tiered access
- Fitur: Free/premium tiers, profile completion, referrals
- Keunikan: Gamified onboarding checklist

Hari 24: Password Reset & Email

Project: MailMagic

- Deskripsi: Email template system untuk password reset
- Fitur: Beautiful email templates, tracking, queue system
- Keunikan: Preview emails sebelum send

Hari 25: Permissions & Groups

Project: AccessControl

- Deskripsi: Permission management dengan role-based access

- Fitur: Custom permissions, group management, audit log
- Keunikan: Permission inheritance visualization

Hari 26: Custom User Model

Project: FlexUser

- Deskripsi: Extended user model dengan phone/username login
- Fitur: Multi-factor authentication, login history
- Keunikan: Biometric login preparation (fingerprint/face)

Hari 27: Session Management

Project: SessionGuard

- Deskripsi: Session security dengan device tracking
- Fitur: Active sessions, force logout, IP whitelist
- Keunikan: Suspicious activity detection

Hari 28: Social Authentication

Project: SocialConnect

- Deskripsi: OAuth integration dengan multiple providers
- Fitur: Google, GitHub, Facebook login, profile sync
- Keunikan: Account linking untuk existing users

Hari 29-30: Review & Integration Project

Project: CommunityHub

- Deskripsi: Mini social network mengintegrasikan konsep bulan 2
- Fitur: User auth, profiles, posts, comments, likes
- Keunikan: Activity feed algorithm, notification system
- Scope: 2 hari untuk build complete MVP